Nazmul Huda Badhon

Deep Learning Researcher
Daffodil International University
Department of Computer Science and Engineering

Email: nazmul.badhon.research@gmail.com
Contact no: +880 190 390 6230
Website, Google Scholar, ResearchGate, LinkedIn

Educational Qualifications

Jan'22 – Dec'25

Bachelor of Science (BSc) in Computer Science and Engineering,
Daffodil International University (DIU)
Thesis: Tabular data-based lightweight convolutional neural network
CGPA: 3.88/4.00 [Top 5% of the cohort]

Sep'23 – Dec'23

Exchange Semester in Computer Engineering, Dongseo University, South Korea
GPA: 4.94/4.50 [Received fully funded GKS-Global Korea Scholarship]

Research Interests

- Deep Learning and Machine Learning
- Vision Language Models
- Sustainable and trustworthy AI

Research Experience

Jan'26 – Present

Graduate Researcher
CSE AI Centre, Daffodil International University

- Developed and organized research datasets for applied artificial intelligence projects.
- Evaluated deep learning experiments for medical imaging and computer vision tasks.
- Prepared research manuscripts for collaborative publications.

Dec'25 – Mar'26

Research Collaborator
8th ABC Challenge, Kyushu Institute of Technology, Kitakyushu, Japan

- Analyzed real-world multimodal sensor data collected from a nursing care facility.
- Developed a Temporal Convolutional Network for sequence modeling.
- Classified indoor locations using BLE beacon signal data from healthcare.

Jan'24 – Dec'25

Student Researcher
Health Informatics Research Lab, Daffodil International University

- Developed and organized research datasets for applied artificial intelligence projects.
- Evaluated deep learning experiments for medical imaging and computer vision tasks.
- Prepared research manuscripts for collaborative publications.

Selected Research Projects

1. Breast Cancer Classification using Hybrid Segmentation Deep Learning
   Developed a hybrid U-Net and CNN model.
   Evaluated model performance using accuracy and ROC-based metrics.

2. Lung and Colon Cancer Histopathology Classification
   Applied preprocessing and transfer learning techniques to histopathology images.
   Evaluated classification performance using accuracy and F1-score.

3. Vision Transformer–Based Medical Image Classification
   Implemented a ViT-based classification pipeline for medical image analysis.
   Achieved 87% test accuracy.

Journal Articles

(* indicates joint co-first authors)

1. **Badhon, N. H.**, Preanto, S. A., Arefin, A. S. S. M. N., Hasan, K. J., & Islam, M. B. (2026). Temporal Convolutional Network Based Indoor Location Recognition from BLE Beacon Signals in Nursing Care Facility. *International Journal of Activity and Behavior Computing*, 2026(2), 1-17.

2. **Badhon, N. H.***, Salehin, I.*, Sajib, M. T. A., Rifat, M. S. H., Noman, S. M., & Moon, N. N. (2025). TLCNN: Tabular data-based lightweight convolutional neural network for electricity energy demand prediction. *Global Energy Interconnection*.

3. Salehin, I., Sajib, M. T. A., **Badhon, N. H.**, Rifat, M. S. H., Amin, N., & Moon, N.N. (2025). Systematic Literature Review of LLM-Large Language Model in Medical: Digital Health, Technology and Applications. *Engineering Reports*, 7(9), e70365.

4. Sajib, M. T. A., **Badhon, N. H.**, Salehin, I., Rifat, M. S. H., Ahmmed, F., & Moon, N. N. (2025). A comparative deep learning methodology for plant insect image classification: assessment of CNN architectures and augmentation techniques. *MethodsX*, 103680.

5. Salehin, I., **Badhon, N. H.**, Sajib, M. T. A., Saifuzzaman, M., & Moon, N. N. (2025). IQ-UltraRecon: Demodulated IQ ultrasound dataset of human hand and arm tissue for deep learning-based reconstruction. *Data in Brief*, 112001.

6. Salehin, I., **Badhon, N. H.**, Sajib, M. T. A., & Moon, N. N. (2025). DeepUS-ReconSeg: A multi-angle paired B-mode Ultrasound dataset for medical imaging reconstruction and segmentation. *Data in Brief*, 112083.

7. Salehin, I., Khan, M. R., Habiba, U., **Badhon, N. H.**, & Moon, N. N. (2024). BAU-Insectv2: An agricultural plant insect dataset for deep learning and biomedical image analysis. Data in Brief, 53, 110083.

Manuscripts under review

1. Islam, A.H.M.S.*, **Badhon, N.H.***, Nur F.N.*, Sajib, M.T.A., Moon, N.N., Sultana, S. (2026). Bridging Neuroscience and Education: A Scoping Review of EEG-Based Cognitive and Emotional States in Educational Settings. Manuscript no: IJEDRO-D-26-00680. *International Journal of Educational Research Open*, Elsevier.

Books & chapters

1. Salehin, I., **Badhon, N. H.**, Sajib, M. T. A., Rifat, M. S. H., & Moon, N. N. (2025). Predicting Peak Energy Demand Using Machine Learning Techniques for Efficient Electricity Supply Management. In *Engineering Applications of AI for Demand Forecasting* (pp. 217-229). CRC Press.

Professional Service

Reviewer Roles

- Peer Reviewer, Digital Health, Sage Publishing, 2025

Honors & Awards

- Research Award for publication in reputed indexed journals, Division of Research and Department of Computer Science and Engineering, 2024-2025
- Best Performer Award, Computer Vision and Medical Image Analysis Bootcamp, DIU, 2025
- University Merit Award, DIU, 2022-2025
- Global Korea Scholarship (GKS) awardee, Government of the Republic of Korea, 2023
- Government General Board Scholarship, Education Board Bangladesh, 2015

Leadership and Extracurricular Activities

- Volunteer Intern, DIU, 2024
- Founding General Secretary, DIU Research Society, DIU, 2024
- Team Leader, International Camp, Dongseo University (DSU), South Korea, 2023
- Event Volunteer, Multi Destination Education Expo, Admission.ac, 2023

Languages

- Bangla: Native
- English: IELTS Academic: Overall 6.5 (Listening: 6.5, Reading: 7, Writing: 7, Speaking: 6)
  Test Date: 4th July 2026

Technical Skills

- Programming: *Python, Java, C*
- Deep Learning: *CNN, U-Net, Vision Transformers, Transfer Learning*
- Frameworks: *PyTorch, TensorFlow, Keras, Scikit-learn*
- Data Analysis: *NumPy, Pandas, Matplotlib, Seaborn*
- Research Tools: *GitHub, Google Colab, LaTeX*
- Version control: *Git*
- Reference Management: *Mendeley*

References

1. **Nazmun Nessa Moon**, Associate Professor
   Department of Computer Science and Engineering
   Daffodil International University, Dhaka, Bangladesh
   Email: moon@daffodilvarsity.edu.bd
   Contact no: +880 179 814 5670

2. **Dr. Fernaz Narin Nur**, Professor
Department of Computer Science and Engineering
Daffodil International University, Dhaka, Bangladesh
Email: fernaz.cse@diu.edu.bd
Contact no: +880 188 641 1541

3. **Imrus Salehin**, PhD Fellow
Department of Computer Science
North Carolina Agricultural and Technical State University
Greensboro, North Carolina, USA
Email: isalehin@aggies.ncat.edu
Contact no: +1 336 255 7549

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